Saramsha Dotel

saramsha.d@gmail.com | (201) 397-8116 | saramsha | dotel-saramsz | New Jersey

Experienced software engineer & technical leader with 6-year track record of planning, designing and building production-grade systems. I specialize in developing AI systems with strengths across backend, data engineering & cloud technologies and have led and scaled technical teams & projects.

KEY SKILLS

- **AI & Data:** Large Language Models (LLMs) • Agentic AI • Retrieval Augmented Generation (RAG) • Prompt Engineering • Predictive Modeling • Data Engineering • MLOps
- **Engineering & Cloud:** System Design • Backend Engineering • DevOps • Cloud Platforms • AWS • Database Management (SQL/NoSQL)
- **Technical Leadership:** People Management • Client Communication • Product Management

EDUCATION & TRAINING

Bachelors in Computer Engineering Dec 2015 - Sep 2019
Institute of Engineering, Tribhuvan University Pulchowk, Nepal

- Graduated with **FIRST** position among all students in the batch with aggregate score of **85.3%**
- Recipient of Ncell **Academic Excellence Award** for the years: 2016, 2017 and 2019

MicroMasters in Artificial Intelligence Apr 2018 - Dec 2020
edX, Columbia University

- Recipient of Fusemachines **AI Fellowship**
- Credentials: CSMM.101x: Artificial Intelligence (AI), CSMM.102x: Machine Learning

RESEARCH PUBLICATIONS

Dotel, S., et.al. (2020). **Disaster Assessment from Satellite Imagery by Analysing Topographical Features Using Deep Learning**. *IVSP20, 86-92.*

PROFESSIONAL EXPERIENCE

Senior Software Engineer Apr 2025 - Present
reAlpha Tech Corp.

- Relocated to the U.S. office of reAlpha following the acquisition of the startup that I co-founded (Naamche) to work closely with stakeholders in marketing, mortgage and real estate departments of the parent company.
- Led the AI transformation of the traditional mortgage brokerage department by building a suite of **Mortgage AI agents** to automate and streamline mortgage workflows in the company. One such agent

automatically parsed the W-2s, pay stubs and bank statements of borrowers and validated it with their loan application reducing the time to verify a loan file from 3 hours to 30 seconds.

- Spearheaded the technical transformation of a **conversational AI platform: AiChat**, reAlpha's subsidiary. Re-architected the system from ground-up to revamp the legacy NLP based product into a LLM-based multi-tenant SaaS platform featuring multi-agent orchestration, voice interactions, digital avatars and improved UI/UX.

Co-founder & Chief Technology Officer Oct 2020 - Present
Naamche Inc (acquired by reAlpha Tech Corp.)

- Co-founded and scaled an AI product development studio from 3 to 40 engineers; drove $0 to $500k ARR, culminating in a **successful acquisition** by a U.S. based company.
- Led the technical development of an AI powered home buying platform for reAlpha overseeing the backend engineering, setting up of data pipelines and deployment of the software solution to AWS cloud. Developed conversational AI assistant **"CLAIRE"** into the platform which was first of its kind AI agent for real estate and mortgage designed to assist home buyers with searching homes, booking tours and negotiating offers.
- Oversaw from ideation to production, the development of **myAlphie**, a product that helped property managers manage apartment turnovers. The product was later successfully sold by the owners to Crawford Hoying for $6 million.
- Architected and engineered the real estate property evaluation engine **"BRAIN"** for reAlpha which leveraged predictive modeling algorithms to score properties on their investment potential by analyzing spatio-temporal and socio-economic data points.
- Spearheaded the development of web and mobile based application for reAlpha which allowed fractional investment in real estate properties with built-in integrations with financial technology platforms.
- Directed the planning, system design, development and delivery of several other software products for global clients:
    - **Agrichat:** a RAG based information retrieval tool developed for the agriculture domain
    - **Atendido:** a POS software for restaurant businesses to manage their operations
    - **Mimir:** a Generative AI based platform for digital cloning; utilized RAG, fine-tuned LLMs and prompt engineering
    - **Sportster:** a social network application for high-school athletes in the US
    - **EthanAI:** a fintech platform for financial portfolio management and analysis
- As the CTO of an early stage startup, set the overall technical vision of the company, led the hiring and mentorship of engineers and designers while fostering good engineering practices and ensuring that customers are satisfied and revenue targets are met.

AI Advisor & Consultant Jan 2025 - Mar 2025
inGrails Pvt. Ltd. Part-time

- Consulted an ed-tech startup building a generative AI powered personalized learning platform. Advised the team on choosing the appropriate technology stack, architecting the RAG pipeline with vector databases and best practices in designing domain-specific AI agents.

Machine Learning Engineer Sep 2019 - Sep 2021
Fusemachines

- Developed an Optical Character Recognition system "FuseExtract" that digitized Nepali and English documents from their photos; tested successfully by a major bank and telecom company in Nepal.
- Built predictive models for Take 2 Interactive based on data mining, natural language processing and time series analysis to forecast the sales and revenue of video games